Evaluation of Bioinformatics Tools for Carbapenem-Resistant Organism Genomic Surveillance and Cluster Detection

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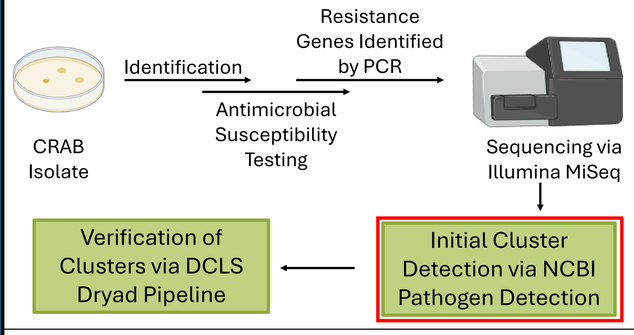
Abstract Carbapenem-resistant organisms (CROs) are microbes that have developed a resistance to carbapenems and are a growing public health threat. The Virginia Division of Consolidated Laboratory Services (DCLS) aims to develop a new, internal pipeline that quickly and accurately identifies genomic clusters for CROs. Towards this goal, we evaluated the ability of clustering methods PopPUNK and floc to accurately assign CRO isolates to previously identified outbreak clusters. We measured the sensitivity and specificity for each method and found that floc maximized sensitivity while maintaining reasonable specificity in comparison to PopPUNK. We concluded that floc is likely the optimal clustering method for carbapenem-resistant organism genomic surveillance and cluster detection at DCLS.

Background

Carbapenems are a class of antibiotics reserved to treat infections caused by multidrug-resistant microbes. Carbapenem-resistant organisms (CROs) are microbes that have developed a resistance to carbapenems and are a growing public health threat. The Virginia Division of Consolidated Laboratory Services (DCLS) is the public health laboratory for the state of Virginia, responsible for performing genomic surveillance for CROs and reporting any identified genomic clusters to the Virginia Department of Health (VDH). Current methods for genomic surveillance and cluster detection of CROs are time intensive and reliant on external resources.

Current Cluster Identification Pipeline



Proposed Change

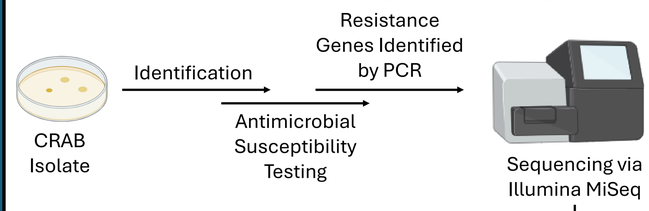
Replace **NCBI Pathogen Detection** with **PopPUNK or floc**

DCLS aims to develop a new, internal pipeline that quickly and accurately identifies genomic clusters. Towards this goal, bioinformatics tools PopPUNK and floc will be tested for their ability to replace NCBI Pathogen Detection as the initial clustering step in the pipeline. Each tool will be evaluated based on its ability to correctly assign each of 674 carbapenem-resistant *Acinetobacter baumannii* to their previously-identified outbreak clusters. Tools will also be evaluated based on speed, output potential, and ease of use.

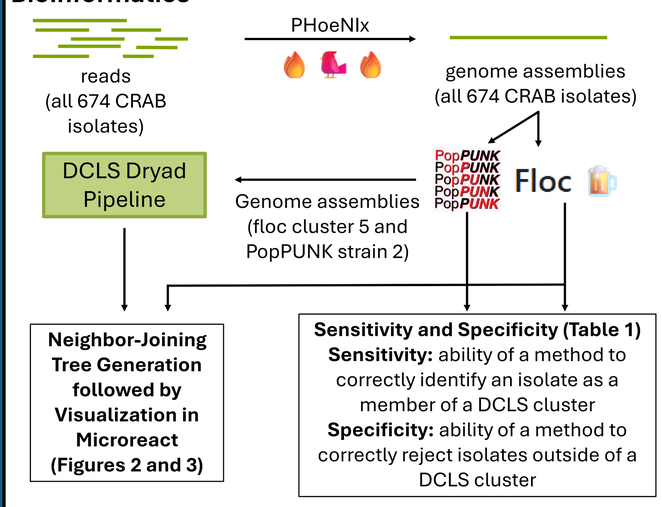
With a new genomic surveillance and cluster detection pipeline that implements the tools tested, DCLS will be able to report CRO surveillance data to VDH more quickly to aid epidemiologic investigations. This improvement will ultimately lead to faster outbreak detection and better-informed public health response, thereby reducing the spread of CROs.

Methods

Sample Acquisition, Processing, Sequencing



Bioinformatics



Results

Table 1. floc and PopPUNK

Main Finding: floc maximizes sensitivity while maintaining specificity

Method	Average Sensitivity	Average Specificity
floc	0.9872	0.8948
PopPUNK	0.9884	0.5523
PopPUNK+PopPIPE	0.7286	0.9782

Table 1. Table depicting average sensitivity and specificity for clustering methods floc and PopPUNK. We attempted two PopPUNK methods, with and without PopPIPE subclustering to see effect on scores.

Figure 2. DCLS Dryad Pipeline vs. floc

Main Finding: floc successfully identifies genetically related isolates

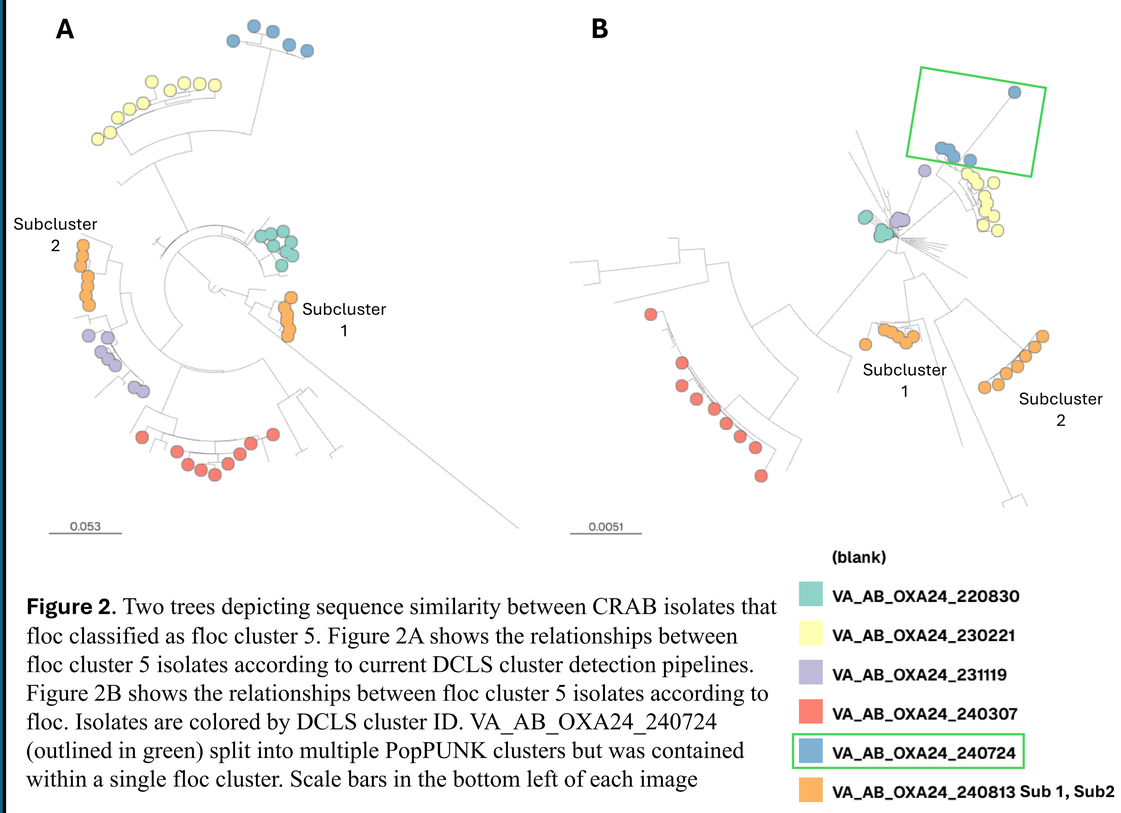


Figure 2. Two trees depicting sequence similarity between CRAB isolates that floc classified as floc cluster 5. Figure 2A shows the relationships between floc cluster 5 isolates according to current DCLS cluster detection pipelines. Figure 2B shows the relationships between floc cluster 5 isolates according to floc. Isolates are colored by DCLS cluster ID. VA_AB_OXA24_240724 (outlined in green) split into multiple PopPUNK clusters but was contained within a single floc cluster. Scale bars in the bottom left of each image

Figure 3. DCLS Dryad Pipeline vs. PopPUNK+PopPIPE

Main Finding: PopPUNK+PopPIPE incorrectly separates genetically related isolates

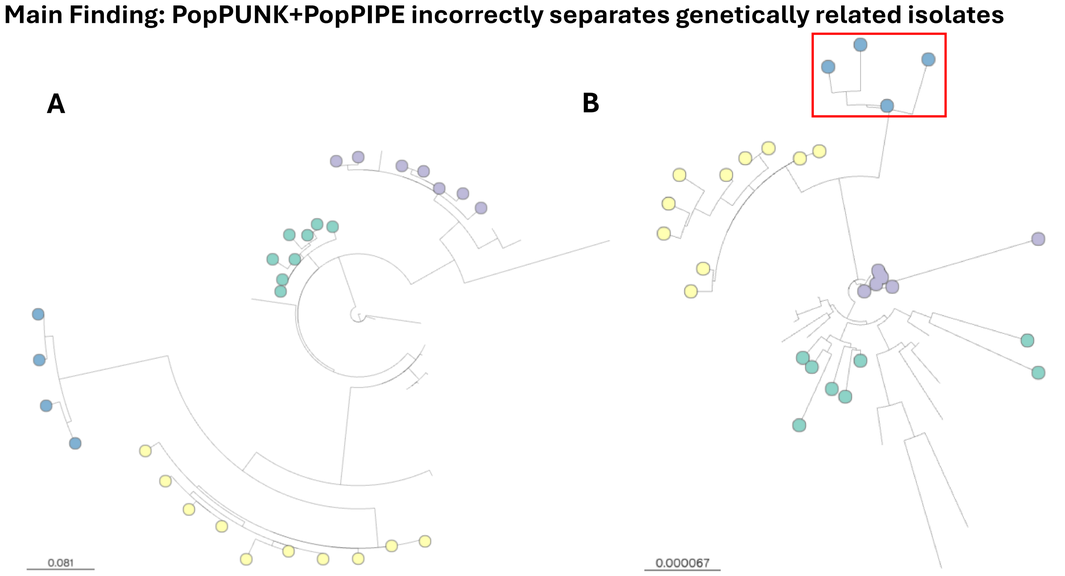


Figure 3. Two trees depicting sequence similarity between CRAB isolates that PopPUNK classified as PopPUNK strain 2. Figure 3A shows the relationships between PopPUNK strain 2 isolates according to current DCLS cluster detection pipelines. Figure 3B shows the relationships between PopPUNK strain 2 isolates according to PopPUNK. Isolates are colored by DCLS cluster ID. VA_AB_OXA24_240724 (outlined in red) split into multiple PopPUNK clusters (4/5 in strain 2, 1/5 in another cluster). Scale bars in the bottom left of each image.

Discussion

CRAB Clustering Output

When comparing the outputs of PopPUNK and floc, we saw a clear tradeoff between sensitivity and specificity. We first compared PopPUNK without PopPIPE subclustering to floc and saw that they were comparably sensitive, but PopPUNK was far less specific (Table 1). Therefore, if we were to follow up floc and PopPUNK with a reference-based SNP calling pipeline, a PopPUNK cluster would be much larger than a floc cluster. This would add time and computational resources to the analyses.

We managed to increase the specificity of PopPUNK by including the PopPIPE subclustering workflow, but sensitivity suffered as a result (Table 1). This meant that PopPUNK+PopPIPE was more likely than floc to incorrectly separate related isolates into multiple PopPUNK clusters, which we observed in Figure 3, where PopPUNK+PopPIPE separated a VA_AB_OXA24_240724 isolate that was only 5-12 SNPs away from all other isolates in VA_AB_OXA24_240724.

Overall, floc was the method that maximized sensitivity while still maintaining reasonable specificity when compared to PopPUNK with and without PopPIPE subclustering.

Preliminary Carbapenem-Resistant *Klebsiella pneumoniae* Clustering Output

We tested floc and both PopPUNK methods on carbapenem-resistant Klebsiella pneumoniae (CRKP). Whereas CRAB in Virginia is highly clonal, CRKP is not. Therefore, we suspected that clustering performance may differ and floc may not be the optimal choice. However, we saw that floc and PopPUNK+PopPIPE both had equal sensitivity and floc had slightly better specificity (Supplementary Table 1). **Therefore, we conclude that floc is also the optimal clustering method for CRKP.**

Next Steps

Floc + Reference-based SNP Calling Pipeline

1. Combine floc with a reference-based SNP calling pipeline to get SNP clusters with smaller SNP ranges and see how the output compares to previous DCLS reports
2. Implement and validate pipeline to DCLS standards

Impact

Faster Reporting to VDH → Faster Outbreak Detection → Reduced Spread of CROs

Supplementary Materials & References

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